5.6 SLN – Structural line

See also: SLNB, SLNP, SLNS, SPT

SLN

Item	Description	Unit	Default
NO	Line number	–	-
NPA	Point at start of line	–	-
NPE	Point at end of line	–	-
REF	Reference to a geometry at GAX	–/*LIT*	-
FIX	Boundary conditions along the line	*Lit*16	-
SDIV	>0: Mesh density used for subdivision	[*m*] 1001	-
	<0: Number of elements to be generated	–	-
GRP	Group number	–	-
STYP	Element type, subdivision	*LIT*	SE
SNO	Number of cross section	–/*Lit*12	-
NP	Number of bore / bedding profile		-
KR	Direction identifier or	*LIT*	-
DRX	explicit direction vector of the local	–	0.
DRY	beam coordinate system	–	0.
DRZ		–	0.
DROT	Additional rotation about beam axis	*deg*	0.
EXA	Eccentricity in global coordinates	[*m*] 1001	0.
EYA	at start	[*m*] 1001	0.
EZA		[*m*] 1001	0.
EXE	Eccentricity in global coordinates	[*m*] 1001	0.
EYE	at end	[*m*] 1001	0.
EZE		[*m*] 1001	0.
FIXA	Hinge conditions at start	–/*Lit*16	-
FIXE	Hinge conditions at end	–/*Lit*16	-
FIMA	Hinge conditions at all interior starts	–/*Lit*16	-
FIME	Hinge conditions at all interior ends	–/*Lit*16	-
XFLG	Intersection properties	*LIT*	-
TITL	Title of Line	*Lit*32	-

This record defines a structural line. Start and end points of the line may be specified explicitly by referencing structural points which have been previously defined at SPT or implicitly using subsequent geometry records of type SLNB / SLNP . In the latter case, it will be checked if a structural point already exists at the respective end coordinate of the line and creates one if not. If no number is given for the structural line at NR, a new number will be assigned automatically.

By entering a negative number at NO the properties of a previously defined structrual line can be changed. For example: SLN -10 SNO 1 changes the cross-section number of line no 10. All other properties of the line remain unchanged. If an additional property record of type SLNS is given when a line is being changed, it will be added to all previously defined property records of the line.

The geometry of a line can be described by adding subsequent records of type SLNB or SLNP. SLNB allows to define straight lines and circular arcs whereas SLNP is used to create freeform curves. A third possibility is to set the geometry by referencing a previously defined axis GAX at REF. If no geometry is defined at all, a straight line between start- and endpoint is assumed.

5.6.1 Local coordinate system

All beam elements which will be created along the structural line have a local coordinate system assigned whose local x-axis is always aligned parallel to the structural line. For curved edges this coordinate system may vary along the line. This local beam-coordinate system determines the orientation of cross sections and, if not otherwise specified at record SLNS, the direction of local supports and kinematic couplings. The orientation of the local z-axis perpendicular to the line can be specified using special literals at KR or directly by entering direction vector at DRX,DRY,DRZ. Following possibilities exist:

- **Explicit direction vector at DRX,DRY,DRZ**

  The entries DRX,DRY,DRZ define a direction vector according to which the local z-axis of the beam coordinate system is aligned. The local x-axis (= tangent) of the coordinate system remains always parallel to the structural line.
  Which of the two local directions y or z is aligned is basically determined by the global variable CTRL LOCA. If nothing is given, the local z-axis is used.

- **Predefined directions using a literal at KR** Instead of an explicit direction, the user may also enter one of the literals POSX, POSY, POSZ, NEGX, NEGY or NEGZ at KR in order to align the local x-axis with one of the given global directions.
  Moreover, the local x-axis may also be aligned within one of the global coordinate planes by entering:

  KR = XY, YX: align x within global XY-plane in X- resp. Y-direction

  KR = YZ, ZY: align x within global YZ-plane in Y- resp. Z-direction

  KR = ZX, XZ: align x within global ZX-plane in Z- resp. X-direction

- **Align to other structural element KR { PT,LN,AR} NO** Using this option, the local z- (or y-) axis of the beam elements along the structural line can be aligned towards another structural element. Following possibilities exist:

  KR PT: align towards a structural point

  KR LN: align towards a structrual line

  KR AR: align towards a structural area

  The direction of each beam element is determined individually by calculating the projection onto the referenced structural item. If no projection exist, i.e. the distance is zero, the